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综述论文

2020年版中国药典总章的发展：综述

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摘要

《中华人民共和国药典》2020年版经国家药品监督管理局和中华人民共和国国家卫生健康委员会于2020年7月审定批准。现行版本于2020年12月30日正式实施。《中国药典》的通则部分讨论了通用的检验方法和指导原则，这些是中国药典药品标准实施的共同要求和基础。由于遵循科学性、通用性、可操作性和可持续发展的原则，2020年版通则相较于以前的版本有所改进。此外，先进成熟分析技术的应用得到了拓展，中药中外源性污染物的检验方法开发得到了加强，技术要求与国际标准的协调性也得到了提升。更新的版本

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1. 介绍

《中国药典》2020年版经国家药品监督管理局（NMPA）和中华人民共和国国家卫生健康委员会审查批准，于2020年7月发布。本版自2020年12月30日起正式实施。《中国药典》是我国药物研制、生产、使用和监管必须执行的法定技术规范。《中国药典》中的通则是准确实施药典的基础。2020年版共收载通则360个，其中新增23个，修订83个。本次修订不仅反映了我国医药行业目前的技术水平，也体现了国际药品质量控制所采用的技术。

2. 高级且成熟的分析技术应用范围已扩大

方法0451，“X射线荧光光谱法”，被新增以指导X射线荧光光谱在元素杂质定性和定量分析中的应用 [1e9]。在方法0601，“相对密度测定”中，新增了振荡换能器密度计的应用和仪器使用 [10,11]。在方法0713，“脂肪和脂肪油的测试”中，熔点范围、皂化值和碘值进行了修订，并新增了有关不皂化物、脂肪酸组成、碱性杂质、过氧化值、甾醇及反式脂肪酸的信息 [12e16]。新增方法1001，“聚合酶链式反应”；1021，“细菌DNA序列鉴定”；及9108，“DNA测序”。这些方法用于确保药物的准确鉴定和临床安全[17e22]。体外方法，即利用仪器测定终点的方法，已取代体内生物方法，这与 t 相符

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Review paper

Development of the general chapters of the Chinese Pharmacopoeia 2020 edition: A review

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ABSTRACT

The Chinese Pharmacopoeia 2020 edition was reviewed and approved by the National Medical Products Administration and the National Health Commission of the People's Republic of China in July 2020. The current edition was officially implemented on December 30, 2020. The general chapters of the Chinese Pharmacopoeia discuss the general testing methods and guidelines, which are the common requirements and basis for the implementation of drug standards in the Chinese Pharmacopoeia. Owing to adherence to the principles of scientificity, versatility, operability, and sustainable development, there is an improvement in the general chapters of the 2020 edition over those of the previous editions. Further, the application of advanced and mature analytical techniques has expanded, the development of testing methods for exogenous pollutants in traditional Chinese medicines has been strengthened, and technical requirements are now better harmonized with international standards. The updated edition provides technical and methodological support to ensure safety, effectiveness, and control of pharmaceuticals in China and will play an important and active role in encouraging the application of advanced technologies, improving the quality control of medicines, and strengthening the means of drug regulation in China. This review provides a comprehensive introduction of the main features of and changes to the general chapters in the Chinese Pharmacopoeia 2020 edition and aims to provide reference for its correct understanding and accurate implementation.

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1. Introduction

The Chinese Pharmacopoeia 2020 edition was reviewed and approved by the National Medical Products Administration (NMPA) and the National Health Commission of the People's Republic of China in July 2020. This edition was officially implemented on December 30, 2020. The Chinese Pharmacopoeia is a statutory technical specification that must be implemented for drug development, production, use, and regulation in China. The general chapters in the Chinese Pharmacopoeia are the basis for the accurate implementation of the Chinese Pharmacopoeia. The 2020 edition contains 360 general chapters, including 23 new and 83 revised chapters. This updated edition reflects not only the current level of technology used in the pharmaceutical industry in China but also the technologies used for international drug quality control.

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2. Application of advanced and mature analytical techniques has been expanded

Method 0451, “X-ray fluorescence spectroscopy”, was added to guide the application of X-ray fluorescence spectroscopy in the qualitative and quantitative analyses of elemental impurities [1–9]. Oscillating transducer density meter application and instrumentation were added in method 0601, “Determination of Relative Density” [10,11]. In method 0713, “Tests of Fat and Fatty Oil”, the melting range, saponification value, and iodine value were revised, and information concerning unsaponifiable matter, fatty acid composition, alkaline impurities, anisidine value, sterols, and trans fatty acids was added [12–16]. Methods 1001, “Polymerase Chain Reactions”; 1021, “Identification of Bacterial DNA Sequences”; and 9108, “DNA Sequencing”, were added. These methods are used to ensure the accurate identification and clinical safety of drugs [17–22]. In vitro methods, which involve the use of an instrument to determine endpoints, have replaced in vivo biological methods, which is in line with the goal of reducing, replacing, and refining laboratory animal use. The anti-factor IIa and anti-factor Xa assays

已在方法1208“肝素生物测定”[23e27]中添加。肝素结合能力测定已添加至方法1213“硫酸鱼精蛋白生物测定”[28e32]。新灭菌方法（蒸汽相灭菌和液相灭菌）已添加到方法1421“灭菌方法”，以指导药品的灭菌/过滤生产，确保无菌水平符合要求[33e36]。单核细胞激活试验已添加到指南9301“注射剂安全性试验应用”[37e42]。基于关于美国药典（USP）和美国分析化学家协会的分析方法验证与转移技术要求的研究，新增了指南9099“药典程序验证”和9100“分析方法转移”。

分析方法的适用性要求得到了加强。在方法1105《非无菌产品微生物检查：微生物计数测试》中，增加了气溶胶测试样品制备的改进方法，并补充了关于小剂量、低含量和小批量样品的测试数量信息。在方法1107《非无菌药品的微生物验收标准》中，对半固体制剂的微生物验收标准进行了修改，以确保其控制严格程度与液体制剂一致。方法1121《抗菌有效性测试》中抗菌防腐剂的定义和范围已被修订，以提高准确性。操作适用性测试中所用培养基和测试微生物的适用性测试回收率也进行了修订，以确保与方法1105[50]的一致性。其他新增加和修订的一般检测方法包括

3. 加强了中药中外源性污染物的检测方法开发

中医药中外源污染物的测定方法在《2020年中国药典》中得到了改进。

表1 中国药典2020版通用检测方法的新增和修订内容

一般章	新增和修订
0421, 拉曼光谱	介绍了透射率、尖端增强拉曼光谱以及成像技术及其在...中的应用
0512, 高效液相色谱 最新的高效液相色谱（HPLC）	物理学、化学、过程控制以及其他分析领域得到了扩展。
0661, 热分析	发展和应用进展已作了全面详细的介绍。有关多维HPLC的信息，添加了电荷气溶胶检测、调整后的色谱条件以及常用定性分析方法。增加了一种热重-质谱法，以实现对……的定性和定量分析
0981, 结晶度	试样中的结晶溶剂（水性）或其他挥发性成分
0991, 特定表面积测定；0992, 固体密度的测定1143, 细菌内毒素测试；9251, 应用指南 细菌内毒素测试	差示扫描量热法被用于测试晶体的尖锐吸热峰的结晶性材料或无定形材料的分散（或无吸热峰）特性。该方法也可以用于在固态的吸热峰位置存在差异时识别晶体形态
1146, 组胺测试；9301, 安全测试应用指南 对于注射	同一化合物的不同晶体形态。
9015, 研究和质量控制指南 药物多晶型	给出了比表面积和固体密度的基本定义和术语。有关的信息提供了仪器和测量方法。
	增加了与国际标准的可追溯性以及假阳性结果和处理方法的描述
	为了避免由于β-葡聚糖引起的误解，凝胶凝块法已被修订，并且最初填充的要求
	变形细胞溶解液，然后加入内毒素的步骤被移除。为了标准化设计、程序和
	干扰实验的限值设置，以确保均匀的质量，并解决变形虫溶细胞素的缺乏问题，
	关于细菌内毒素含量的信息，用于限值设定、方法选择和预处理
	已添加用于测试样品的方法。重组因子C测定适用于检测含有...的样品
	引入了β-葡聚糖、B因子和凝血酶原，以解决阿米巴细胞裂解液资源短缺的问题。
	测试组胶溶液的制备方法、方法适用性测试及其测定
	已添加最小有效浓度测试。
	添加了固态核磁共振光谱方法。化学环境的差异
	测试样品的不同晶体形式的相同原子核会导致化学位移和耦合常数的差异，
	以及在识别晶体状态时的相对强度。

were added to method 1208, “Biological Assay of Heparin” [23–27]. The heparin-binding capacity assay was added to method 1213, “Biological Assay of Protamine Sulfate” [28–32]. New sterilization methods (vapor-phase sterilization and liquid-phase sterilization) were added to method 1421, “Methods of Sterilization”, to guide the sterilization/filtration-based production of drugs to ensure that the sterility level meets the requirements [33–36]. The monocyte activation test was added in guideline 9301, “Application of Safety Tests for Injection” [37–42]. Based on research detailing the technical requirements for the verification and transfer of analytical methods from the United States Pharmacopoeia (USP) and the American Association of Analytical Chemists [43–49] to the Chinese Pharmacopoeia, guidelines 9099, “Verification of Compendial Procedures”, and 9100, “Transfer of Analytical Procedures”, were added.

The applicability requirements of the analytical methods were enhanced. In method 1105, “Microbiological Examination of Non-sterile Products: Microbial Enumeration Tests”, an improved method for the preparation of aerosol test samples was included, and information regarding the quantities of small-dose, low-content, and small-batch samples to be tested was added. In method 1107, “Microbiological Acceptance Criteria of Nonsterile Pharmaceutical Products”, the microbiological acceptance criteria for semisolid preparations were modified to ensure strict control, as is required for liquid preparations. The definition and scope of antimicrobial preservatives in method 1121, “Antimicrobial Effectiveness Testing”, have been revised to enhance accuracy. The recovery rates of the suitability test for the medium and test microorganisms used in the operational suitability test were revised to ensure consistency with method 1105 [50]. Other newly added and revised general testing methods are detailed in Table 1.

3. Development of testing methods for exogenous pollutants in traditional Chinese medicines (TCMs) has been strengthened

Methods for determining exogenous pollutants in TCMs have been improved in the Chinese Pharmacopoeia 2020 edition.

Table 1 Additions and revisions in the Chinese Pharmacopoeia 2020 edition general testing methods.

General chapter	Additions and revisions
0421, Raman Spectroscopy	Transmittance, tip-enhanced Raman spectroscopy, and imaging techniques were introduced, and their applications in the fields of physics, chemistry, process control, and other analyses was expanded.
0512, High-Performance Liquid Chromatography	The latest HPLC developments and application progress were fully detailed. Information about multidimensional HPLC, charged aerosol detection, adjusted chromatographic conditions, and common qualitative analysis methods was added.
0661, Thermal Analysis	A thermogravimetry-mass spectrometry method was added to realize the qualitative and quantitative analyses of crystallization solvents (aqueous) or other volatile components in a test sample
0981, Crystallinity	Differential scanning calorimetry was added for crystallinity tests of the sharp endothermic peak of crystalline materials or the dispersion (or no endothermic peak) characteristics of amorphous materials. This method can also be used to identify the crystalline form when there is a difference in the endothermic peak position of the solid state of different crystal forms of the same compound.
0991, Determination of Specific Surface Area; 0992, Determination of the Density of Solids 1143, Test for Bacterial Endotoxins; 9251, Guideline for the Application of the Bacterial Endotoxin test	The basic definitions and terminology of specific surface area and solid density are given. Information about instruments and measuring methods is provided. Traceability with international standards and a description of false-positive results and processing methods were added to avoid misinterpretation due to β-glucans. The gel-clot method was revised, and the requirement to initially fill the amoebocyte lysate, followed by the addition of endotoxin, was removed. To standardize the design, procedure, and limit setting of interference experiments, to ensure even quality, and to address a lack of amoebocyte lysate, information about the contents of the bacterial endotoxin for limit setting, choice of methods, and pretreatment methods for test samples were added. The recombinant factor C assay, which is suitable for testing samples containing β-glucans, factor B, and prothrombin, was introduced to address the shortage of amoebocyte lysate resources. The preparation method for the test histamine solution, the method suitability test, and the determination of the minimum valid concentration test were added.
1146, Test for Histamines; 9301, Guideline for the Application of Safety Tests for Injections	
9015, Guideline for Studies and Quality Control of Drug Polymorphisms	Solid nuclear magnetic resonance spectroscopy method was added. The differences in the chemical environment of the same atomic nucleus of different crystal forms of a test sample causes differences in chemical shifts, coupling constants, and relative intensities during identification of the crystalline states.

Qualitative screening and quantitative analytical methods, including gas chromatography-tandem mass spectrometry (GC-MS/MS) and high-performance liquid chromatography-tandem mass spectrometry (HPLC-MS/MS), were added to method 2341, “Determination of Pesticide Residues”. Qualitative screening methods are used for the rapid qualitative testing, risk monitoring, and early warning testing of pesticides. Pesticides with limited requirements can be directly determined using a quantitative analytical method. In the 2020 edition, the number of pesticides tested has increased to 592. Eighty-eight pesticides were identified using GC-MS/MS and 523 pesticides were determined using HPLC-MS/MS. For pesticides that can be determined with GC-MS/MS and HPLC-MS/MS, the preferred method is provided, and the maximum possible number of characteristic ions is recommended [51–56].

Accurate and low-detection-limit methods, including HPLC-MS/MS analysis of aflatoxin and patulin; HPLC and HPLC-MS/MS analyses of ochratoxin A, zearalenone, and vomitoxin; and HPLC-MS/MS analysis of multiple mycotoxins, were added to method 2351, “Determination of Mycotoxins”. Considering that the above-mentioned methods require complex sample pretreatment, specific instruments, and specialized personnel training, they will be subject to limitations for being used in quality control of TCMs. A fast, sensitive, simple, and low-cost aflatoxin ELISA method was added as a new technique for the quality control of TCM in China (Table 2) [57–63].

As the development of acceptable microbiological criteria for TCM decoction pieces was a breakthrough, strategies and methods for specific microbial contamination control in TCMs with different uses were introduced. Compared with the medicines produced according to good manufacturing practices, TCM decoction pieces contain microorganisms in high abundance, with a wider variety of species and a more uneven distribution. Further, the microbial analysis requirements are unique to each type of medicinal material. Therefore, method 1108, “Microbiological Examination of Traditional Chinese Medicine Decoction Pieces”, was added. The enumerated microbial parameters include total aerobic microbial counts, total combined yeast/mold counts, and the number of heat-resistant bacteria; the parameters for specified microorganisms

表2 黄曲霉素检测的免疫学方法与化学方法比较。

Item	预处理	灵敏度类型 回收率 (%)					设备	速度成本 (人民币/ 测试)	人员	参考文献
		大麦果发芽	枣子 刺荆 精液	桃 精液	薏苡 精液					
ELISA	直接稀释或提取, 20至60分钟	纳克/毫升 AFB ₁ 82.8e95.9 AFTs 98.4至110.4	74.7e88.5 87.0e98.6	94.1至101.9 104.0e112.0	84.0e89.1 86.8e103.3	荧光光度计 (10,000至30,000元人民币)	60 min/ test	30e50	普通人员 可以操作	[58]
	高效液相色谱免疫亲和柱, 4e8 h	纳克/毫升 AFB ₁ 61.3e70.9 AFTs 62.5e78.5	57.0e61.8 58.1e67.8	68.1e78.5 69.2e84.0	63.9到69.7 65.1e74.5	高效液相色谱 (<100, 000元)	3 h/ test	200—300	需要训练 运营人员	[58]

AFB1: 黄曲霉素B1; AFTs: 总黄曲霉素。

包括耐胆汁革兰氏阴性菌、大肠杆菌和沙门氏菌的数量。规定了待测试产品的数量、测试溶液的制备方法以及计数方法的适用性测试；对于中药材来说，结果解释的不确定性可能比其他产品更大。

在方法2322“汞和砷的形态及价态的测定”中，改进了测试溶液的制备方法，以应对海洋和动物来源中药材中砷和汞价态测定的困难。特别是，有关制备的信息

添加了测试溶液、样品量的确定及方法应用原理。

4. 技术要求与国际人用药品注册技术协调委员会（ICH）的技术要求指南更好地协调一致

2017年，国家药品监督管理局（NMPA）加入了国际协调会议（ICH）。在编纂《中华人民共和国药典》2020年版的过程中，国际标准的实施进一步得到加强（表3）。考虑到当前药品生产的状况和

include the number of bile-tolerant gram-negative bacteria, *Escherichia coli*, and *Salmonella*. The quantity of the product to be tested, the preparation method for the test solution, and the suitability test of the counting method are specified; and the uncertainty in the interpretation of the results can be greater for TCMs than that for other products. In method 2322, “Determination of Mercury and Arsenic Speciation and Valence States”, the method of test solution preparation was improved to address the difficulties in determining the valence states of arsenic and mercury in marine- and animal-derived TCMs. Notably, information regarding the preparation of

the test solution, determination of the sample amount, and principle of the method application, was added.

4. Technical requirements are better harmonized with the International Council for Harmonization (ICH) of technical requirements for pharmaceuticals for human use guidelines

In 2017, the NMPA joined the ICH. In the process of compiling the Chinese Pharmacopoeia 2020 edition, the implementation of international standards was further strengthened (Table 3) [64–67]. Considering the current status of drug production and

表3 ICH Q4B 在《2020年中国药典》中的实施情况。

ICH 编号/中文药典 No.	测试方法	实施状态	主要区别	参考文献
附件 1/0841	燃烧残渣/硫酸盐灰	在……过程中实施	硫酸加入量、点火温度，以及实验结束的条件	[64e66]
附录 2/0102 0942	静脉制剂可提取体积的测试过程	实施	抽样方法、方法细节和结果解读	[64e66]
附件 3/0903	颗粒污染测试：过程中不可见颗粒	实施	光遮挡颗粒计数仪器校准测试，环境水样品的检测要求，抽样方法，以及对带标记注射的评估	[64e66]
附录4A/1105	非无菌产品的微生物检测：微生物计数测试	在……过程中实施	菌株、培养基和方法详情	[64,66]
附录 4B/1106	非无菌产品的微生物学检验：指定微生物检测	正在进行实施	菌株、培养基、方法详情及结果解读	[64,66]
附录 4C/1107	非无菌产品的微生物学检验：药品制剂的验收标准	在……过程中实施	范围、沙门氏菌检测及微生物验收标准	[64,66]
附件 5/0921	供药用的物质崩解试验	在……过程中实施	小剂量和微剂量制剂，如贴片，以及标准用于传统中药（植物药）设备，结果解释	[64e66]
附件 6/0941	剂量单位的一致性	在……过程中实施	方法，结果解读	[64e66]
附录7/0931	溶出度测试	在……过程中实施	方法，结果解读	[64e66]
附件 8/1101	无菌检测	在……过程中实施	菌株、待测试产品数量、过滤次数和数量	[64,66]
附件 9/0923	片剂脆碎度	在……过程中实施	冲洗液的仪器，笔记	[64e67]
附件10/0541	聚丙烯酰胺凝胶电泳	在……过程中实施	方法详情	[64e66]
附录11/0542	毛细管电泳	在……过程中实施	方法详情	[64e66]
附件12/0982	分析筛分	在……过程中实施	中国药典包括手工筛分法;该 ICH 指南包括声波筛分法	[64e66]
附件13/0993	粉末的堆积密度和轻敲密度	在……过程中实施	无变化	[64e66]
附录14/1145	细菌内毒素测试	在……过程中实施	方法描述	[64,66]

Table 3 Implementation status of the ICH Q4B in the Chinese Pharmacopoeia 2020 edition.

ICH No./Chinese Pharmacopoeia No.	Testing method	Implementation status	Main differences	Refs.
Annex 1/0841	Residue on ignition/sulfated ash	In the process of implementation	Sulfuric acid addition amount, ignition temperature, and conditions for the end of the experiment	[64–66]
Annex 2/0102, 0942	Test for extractable volume of parenteral preparations	In the process of implementation	Sampling method, method details, and result interpretation	[64–66]
Annex 3/0903	Test for particulate contamination: subvisible particles	In the process of implementation	Instrument calibration for the light obscuration particle count test, requirements for testing environmental water samples, sampling method, and the evaluation of injections with a labeled volume of 100 mL	[64–66]
Annex 4A/1105	Microbiological examination of nonsterile products: microbial enumeration tests	In the process of implementation	Strains, medium, and method details	[64,66]
Annex 4B/1106	Microbiological examination of nonsterile products: test for specified microorganisms	In the process of implementation	Strains, medium, method details, and result interpretation	[64,66]
Annex 4C/1107	Microbiological examination of nonsterile products: acceptance criteria for pharmaceutical preparations and substances for pharmaceutical use	In the process of implementation	Scope, <i>Salmonella</i> tests, and microbial acceptance criteria for small and microdose preparations such as patches, and standards for traditional Chinese medicines (vegetable medicines)	[64,66]
Annex 5/0921	Disintegration test	In the process of implementation	Apparatus, result interpretation	[64–66]
Annex 6/0941	Uniformity of dosage units	In the process of implementation	Methods, result interpretation	[64–66]
Annex 7/0931	Dissolution test	In the process of implementation	Methods, result interpretation	[64–66]
Annex 8/1101	Sterility test	In the process of implementation	Strains, number of products to be tested, filter times, and quantity of the rinsing fluid	[64,66]
Annex 9/0923	Tablet friability	In the process of implementation	Apparatus, notes	[64–67]
Annex 10/0541	Polyacrylamide gel electrophoresis	In the process of implementation	Method details	[64–66]
Annex 11/0542	Capillary electrophoresis	In the process of implementation	Method details	[64–66]
Annex 12/0982	Analytical sieving	In the process of implementation	Chinese Pharmacopoeia includes the manual sieving method; the ICH guideline includes the sonic-sifter sieving method	[64–66]
Annex 13/0993	Bulk density and tapped density of powders	In the process of implementation	No changes	[64–66]
Annex 14/1145	Bacterial endotoxins test	In the process of implementation	Method description	[64,66]

对市场上现有产品的质量控制及适用性, 新增加的一般技术要求与ICH指南保持一致, 修订后的一般技术要求尽可能与ICH指南更好地协调。

稳定性是影响药品市场竞争力的关键因素之一, 也是技术创新的重要领域。指南9001《药物原料及制剂稳定性试验》已根据ICH Q1A [68]进行了修订。提出了制剂质量“显著变化”的定义, 以指导生产企业关注关键质量属性。此外, 对温度敏感药物的运输要求也进行了明确说明。对于特殊制剂, 如缓释和控释制剂及吸入剂, 列出了影响其稳定性试验的重要参数。指南9101《分析方法验证》已修订, 以与ICH Q2 [69]保持一致; 删除了关于修正因子的内容, 并修订了准确度和精密度的方法。同时还涉及药物杂质的报告、鉴定及鉴定阈值, 以及IC的决策流程。

5. 总结与展望

《中国药典》2020版的通则章节以科学、风险和适用性为基础, 并参考了ICH指南。近年来发展起来的新技术和新要求被引入, 以提供技术 and 方法支持, 确保中国药品的安全性、有效性和可控性。本版将积极推动先进技术的应用, 提升药品质量控制水平, 并加强中国药品监管手段。

从其他药典的历史可以看出, 由于科学认知的局限, 药品标准的发展是一个逐步持续改进的过程。按质量设计和生命周期管理的理念将在2025年版《中国药典》通则中进一步实施。例如, 将引入分析程序生命周期指南和工艺分析技术, 并加强统计方法在分析方法的开发、验证、转移和确认中数据评估、解释和处理的作用。基于风险评估的微生物控制体系也将得到改进。

广泛使用的分析技术, 如高效液相色谱 (HPLC)、气相色谱 (GC) 和原子光谱将进行修订。此外, 还将适时引入更多科学、客观和便捷的技术。针对特定制剂的个性化微生物检测方法的开发以及快速微生物学方法的研究将得到进一步阐述。对中药粗药中有效成分和毒性成分、外源性污染物的检测方法, 以及中药饮片的微生物学检查要求将继续完善。

2020年, ICH启动了Q4B指南的修订工作。中国药典委员会 (ChPC) 将继续扩大其在药品标准协调化方面的参与, 并在验证基础上积极促进与ICH Q4的协调/互换性。其他ICH指南, 如ICH Q3D, 也将实现协调。此外, 中国药典中“元素杂质限度及程序”的通用章节将被制定, 以更好地评估和控制中国药品中的元素杂质。

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参考文献

[1] X.X. 许, Z. 刘, X.X. 洪, X射线荧光光谱在药物元素杂质分析中的应用, 药物与临床研究, 27 (2019) 368-370. [2] M. Resano, M.R. Flórez, I. Queralt, 等, 使用高分辨率连续源石墨炉原子吸收光谱法及直接固体样品分析测定废汽车催化剂和活性药物成分中的钯、铂和铑, 光谱学报B: 原子光谱学, 105 (2015) 38-46. [3] N. Lewen, M. Soumeillant, J. Qiu, 等, 使用便携式XRF仪器促进金属催化剂捕捉剂筛选, 有机工艺研究与开发, 19 (2015) 2039-2044. [4] D. Davis, H. Furukawa, 将XRF作为血浆光谱化学的替代技术, 以应对药物材料中元素杂质的新USP和ICH指导, 光谱学, 32 (2017) 12-17. [5] V. Balaram, 元素杂质测定的最新进展

quality control and the current applicability of products already on the market in China, the newly added general technical requirements are consistent with the ICH guidelines, and the revised general technical requirements are better harmonized with the ICH guidelines as much as possible.

Stability is one of the critical factors influencing competitiveness in the drug market and is an important field of technological innovation. Guideline 9001, “Stability Testing of Drug Substances and Preparations”, was revised according to the ICH Q1A [68]. The definition of “significant changes” in preparation quality was proposed to guide manufacturers to focus on critical quality attributes. Additionally, the requirements for the transportation of temperature-sensitive drugs have been clarified. For special preparations, such as sustained- and controlled-release preparations and inhalations, the important parameters affecting their stability test are listed. Guideline 9101, “Validation of Analytical Methods”, was revised to be consistent with the ICH Q2 [69]; the contents regarding the correction factor were deleted and the methods for accuracy and precision were revised. The reporting, identification, and qualification thresholds for drug impurities and the decision tree of the ICH Q3A and Q3B [70,71] were introduced in guideline 9102, “Analysis of Impurities in Drugs”. To ensure consistency with the ICH Q3C [72], cumene and methyl isobutyl ketone were revised from class 3 solvents to class 2 solvents, and triethylamine, a class 3 solvent, was added to method 0861, “Determination of Residual Solvents”. The flow-through cell and reciprocating cylinder methods were added to method 0931, “Dissolution and Drug Release Test”. The instruments, methods, and interpretations related to the ICH Q4B Annex 7 [73–82] and the research results on specific preparations, such as compound ketoconazole cream and lithium carbonate sustained-release tablets, were introduced. Method 1101, “Sterility Tests”, was improved, based on the ICH Q4B Annex 8, to be more instructive and practical [83]. The scope of environmental monitoring, the storage and use of culture media and strains, the culture time of the medium sensitivity test, the number and quantity of products to be tested, and the requirements of incubation and observation were revised. The bulk density and tapped density are important functionality-related characteristics of pharmaceutical excipients in powder form. These densities are commonly used to calculate the Hausner ratio and compressibility index of the powders. Referring to the ICH Q4B Annex 13 [84–86], method 0993, “Bulk Density and Tapped Density of Powders”, was added. Referring to the ICH M7 [87], guideline 9306, “Genotoxic Impurities Control”, was added, and the general principles, assessment methods, and calculation methods of acceptable intakes and limits were introduced.

However, some general testing methods in the Chinese Pharmacopoeia 2020 edition still differ from those in the ICH Q4B (Table 3). The general testing methods in the Chinese Pharmacopoeia were originally drafted according to the British Pharmacopoeia and the World Health Organization, and these general chapter methods have a long history of use and a wide variety of applications in China. However, the current mainstream drug standard harmonization is based on the Pharmacopoeia Discussion Group and the ICH. Moreover, due to limited information and the complexity of regulatory adjustments, the information in international standards referenced by the Chinese Pharmacopoeia is not comprehensive, and the revisions are not timely. Despite these challenges, harmonization with international standards is still vigorously promoted by the Chinese Pharmacopoeia Commission (ChPC). In October 2018, the ICH Q4 symposium was held in Beijing. More than 20 experts from the ICH Expert Working Group and ChPC discussed strategies for implementing ICH Q4 in China. In 2020, the ICH Q4B implementation status of the Chinese Pharmacopoeia was added to the official ICH website for the first time [64].

5. Summary and prospects

The general chapters of the Chinese Pharmacopoeia 2020 edition are based on science, risk, and applicability, and refer to the ICH guidelines. New technologies and requirements developed in recent years were introduced to provide technical and methodological support to ensure the safety, effectiveness, and controllability of pharmaceuticals in China. The current edition will play an active role in encouraging the application of advanced technologies, improving quality control of drugs, and strengthening the means of drug regulation in China.

As observed from the history of other pharmacopoeias, the development of drug standards is a process of gradual and continuous improvement owing to the limitations of scientific cognition. The concept of quality by design and life cycle management will be further implemented in the general chapters of the 2025 Chinese Pharmacopoeia [88–96]. For example, analytical procedure lifecycle guidelines and process analysis technologies will be introduced, and the roles of statistical methods in data evaluation, interpretation, and processing for the development, validation, transfer, and verification of analytical methods will be strengthened. The system for microbiological control based on risk assessment will also be improved.

Widely used analytical technologies, such as HPLC, GC, and atomic spectroscopy, will be revised. Moreover, additional scientific, objective, and convenient techniques will be introduced in a timely manner. The development of personalized microbial testing methods for specific preparations and research detailing rapid microbiological methods will be further elaborated [97–101]. The testing methods for active and toxic ingredients, exogenous pollutants in crude TCMs, and the microbiological examination requirements for TCM decoction pieces will continue to be improved.

In 2020, the revision of the Q4B guidelines was initiated by the ICH. The ChPC will continue to expand its participation in the harmonization of drug standards and actively promote harmonization/interchangeability with ICH Q4 based on validation. Other ICH guidelines, such as the ICH Q3D, will also be harmonized. Further, the general chapter of “Elemental Impurities Limits and Procedures” in the Chinese Pharmacopoeia will be developed to better assess and control elemental impurities in drugs in China.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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References

[1] X.X. Xu, Z. Liu, X.X. Hong, Application of X-ray fluorescence spectrometry in analysis of drug elemental impurities, Pharm. Clin. Res. 27 (2019) 368–370. [2] M. Resano, M.R. Flórez, I. Queralt, et al., Determination of palladium, platinum and rhodium in used automobile catalysts and active pharmaceutical ingredients using high-resolution continuum source graphite furnace atomic absorption spectrometry and direct solid sample analysis, Spectrochim. Acta Part B: At. Spectrosc. 105 (2015) 38–46. [3] N. Lewen, M. Soumeillant, J. Qiu, et al., Use of a field-portable XRF instrument to facilitate metal catalyst scavenger screening, Org. Process Res. Dev. 19 (2015) 2039–2044. [4] D. Davis, H. Furukawa, Using XRF as an alternative technique to plasma spectrochemistry for the new USP and ICH directives on elemental impurities in pharmaceutical materials, spectroscopy 32 (2017) 12–17. [5] V. Balaram, Recent advances in the determination of elemental impurities in

制药——现状、挑战与前沿趋势，《分析化学趋势》80 (2016) 83-95。[6] H. Furukawa, N. Ichimaru, K. Suzuki 等，《药物原料中杂质分析的校准曲线法与背景基本参数法的比较验证》，《X 射线光谱学》46 (2017) 382-387。[7] 美国药典公约，<735> X 射线荧光光谱法，收录于《美国药典》第 41 册，第 4 卷，United Book Press, 巴尔的摩，2018 年，第 6486-6491 页。[8] 欧洲药典委员会，2.2.37 X 射线荧光光谱法，收录于《欧洲药典》第十版，Druckerei C.H. Beck, 北林根，2020 年，第 65-66 页。英国药典委员会，附录 II F: X 射线荧光光谱法，收录于《英国药典 2021》，第 V 卷，英国版政务署，伦敦，2020 年，V-A198-V-A199。[9] 欧洲药典委员会，2.2.5 相对密度，收录于《欧洲药典》第十版，Druckerei C.H. Beck

硫酸蛋白胨，秦。《药学杂志》杂志。8 (2017) 1541e1547。[30] 日本药典委员会，硫酸蛋白胨，载于《日本药典》第17版，厚生劳动省，东京，2016年，第1483e1484页。[31] 《美国药典公约》，硫酸蛋白胨，载于：《美国药典》41卷第3卷，联合图书出版社，巴尔的摩，2018年，第3503e3504页。[32] 欧洲药典委员会，硫酸蛋白胨，载于：欧洲药典第10版，Druckerei C.H. Beck, N'ordlingen，2020年，第3667e3669页。[33] 美国药典公约，<1229. 1> 直接接触蒸汽灭菌，<1229. 2> 水性液体的湿热灭菌，<1229. 4> 液体消毒过滤，<1229. 6> 液相灭菌，<1229. 11> 蒸汽相灭菌，收录于：美国药典41、7716、7719 5、2018年、7689 7706、7709 7722、7733 7734卷，联合图书出版社，巴尔的摩，页 e - -。[34] 欧洲药典 Commi

pharmaceuticals-Status, challenges and moving frontiers, Trends Anal. Chem. 80 (2016) 83–95.

[6] H. Furukawa, N. Ichimaru, K. Suzuki, et al., The comparative verification of calibration curve and background fundamental parameter methods for impurity analysis in drug materials, X Ray Spectrom. 46 (2017) 382–387.

[7] The United States Pharmacopeial Convention, <735> X-ray fluorescence spectrometry, in: United States Pharmacopeia 41, Vol. 4, United Book Press, Baltimore, 2018, pp. 6486–6491.

[8] The European Pharmacopoeia Commission, 2.2.37, X-ray fluorescence spectrometry, in: European Pharmacopoeia 10th Edition, Druckerei C.H. Beck, Nördlingen, 2020, pp. 65–66.

[9] The British Pharmacopoeia Commission, Appendix II F: X-ray fluorescence spectrometry, in: British Pharmacopoeia 2021, Vol. V, Her/His Majesty's Stationary Office, London, 2020. V-A198-V-A199.

[10] The European Pharmacopoeia Commission, 2.2.5 Relative density, in: European Pharmacopoeia 10th Edition, Druckerei C.H. Beck, Nördlingen, 2020, pp. 25–26.

[11] W. Saburo, O. Atsusi, S. Yasutaka, Amakara, A comparison between the hydrometer method and the oscillating-type density meter method in ethanol concentration measurement, J. Brew. Soc. Jpn. 102 (2007) 155–159.

[12] The United States Pharmacopeial Convention, <401> Fats and fixed oil, in: United States Pharmacopeia 41, Vol. 4, United Book Press, Baltimore, 2018, pp. 6184–6197.

[13] Animal and vegetable fats and oils — gas chromatography of fatty acid methyl esters — Part 1: Guidelines on modern gas chromatography of fatty acid methyl esters, ISO International Standard 12966–1 (2014).

[14] Fat (total, saturated and unsaturated) in foods, hydrolytic extraction gas chromatographic method, AOAC Official Method 6 (2001), 996.

[15] M.A. P Muniz, M.N.F.D. Santos, C.E.F. Costa, et al., Physicochemical characterization, fatty acid composition, and thermal analysis of Bertholletia excelsa HBK oil, Phcog. Mag. 11 (2015) 147–151.

[16] E.N. Anderson-Foster, A.S. Adebayo, N. Justiz-Smith, Physico-chemical properties of Blighia sapida (ackee) oil extract and its potential application as emulsion base, Afr. J. Pharm. Pharmacol. 6 (2012) 200–210.

[17] The United States Pharmacopeial Convention, <1113> Microbial Characterization, Identification, and Strain Typing; <1125>Nucleic acid-based techniques; <1126> Nucleic acid-based techniques-extraction, detection and sequencing; <1127> Nucleic acid-based techniques-amplification; <1128> Nucleic acid-based techniques-microarray; <1129> Nucleic acid-based techniques-genotyping; <1130> Nucleic acid-based techniques-approaches for detecting trace nucleic acids (residual DNA testing); <1132> Residual Host Cell Protein Measurement in Biopharmaceuticals, in: United States Pharmacopeia 41, Vol. 5, United Book Press, Baltimore, 2018, pp. 7301–7305, 7353-7414.

[18] The European Pharmacopoeia Commission, 2.6.21 Nucleic acid amplification techniques, 2.6.7 Mycoplasma, in: European Pharmacopoeia 10th Edition, Druckerei C.H. Beck, Nördlingen, 2020, pp. 219–224, 194-199.

[19] The Committee on Japanese Pharmacopoeia, G4 Microorganisms: rapid identification of microorganisms based on molecular biological methods, G5 Purity test on crude drugs using genetic information, in: Japanese Pharmacopoeia 17th Edition, Ministry of Health, Labour and Welfare, Tokyo, 2016, pp. 2503–2505, 2516-2519.

[20] G. Xu, X. Wang, C. Liu, et al., Authentication of official Da-huang by sequencing and multiplex allele-specific PCR of a short maturase K gene, Genome 56 (2013) 109–113.

[21] Y. Yuan, Z.Q. Wang, C. Jiang, et al., Establishment of polymerase chain reaction method in Chinese Pharmacopoeia (2020 edition), China J. Chin. Mater. Med. 45 (2020) 4537–4544.

[22] C. Jiang, Y. Yuan, M.F. Chen, et al., Molecular authentication of multispecies honeysuckle tablets, Genet. Mol. Res. 12 (2013) 4827–4835.

[23] The United States Pharmacopeial Convention, <208> Anti-factor Xa and anti-factor IIa assays for unfractionated and low molecular weight heparins, in: United States Pharmacopeia 41, Vol. 4, United Book Press, Baltimore, 2018, pp. 6113–6117.

[24] The European Pharmacopoeia Commission, 2.7.5 Assay of heparin, in: European Pharmacopoeia 10th Edition, Druckerei C.H. Beck, Nördlingen, 2020, pp. 269.

[25] C. Martinez, A. Savadogo, C. Agut, et al., Reproducibility of the anti-Factor Xa and anti-Factor IIa assays applied to enoxaparin solution, J. Pharmaceut. Biomed. Anal. 81–82 (2013) 138–145.

[26] T. Suzuki, A. Ishii-Watabe, N. Hashii, et al., The establishment and validation of efficient assays for anti-IIa and anti-Xa activities of heparin sodium and heparin calcium, Biologicals 41 (2013) 415–423.

[27] A.L. Berkovskii, E.V. Sergeeva, A.V. Suvorov, et al., Evaluating the activity of low-molecular-weight heparin in preparations and substances, Pharm. Chem. J. 46 (2012) 249–252.

[28] Y. Zhang, Z. Li, D.J. Tan, et al., Feasibility study of heparin-binding capacity method for testing biological potency of protamine sulfate, Chin. J. Pharm. Anal. 37 (2017) 1260–1265.

[29] Y.D. Guo, B. Wu, Y.C. Hu, et al., Study on the standard of biological assay of

protamine sulfate, Chin. J. Pharm. Anal. 8 (2017) 1541–1547.

[30] The committee on Japanese pharmacopoeia, protamine sulfate, in: Japanese Pharmacopoeia 17th Edition, Ministry of Health, Labour and Welfare, Tokyo, 2016, pp. 1483–1484.

[31] The United States Pharmacopeial Convention, Protamine sulfate, in: United States Pharmacopeia 41, Vol. 3, United Book Press, Baltimore, 2018, pp. 3503–3504.

[32] The European Pharmacopoeia Commission, Protamine sulfate, in: European Pharmacopoeia 10th Edition, Druckerei C.H. Beck, Nördlingen, 2020, pp. 3667–3669.

[33] The United States Pharmacopeial Convention, <1229.1> Steam sterilization by direct contact, <1229.2> Moist heat sterilization of aqueous liquids, <1229.4> Sterilizing filtration of liquids, <1229.6> Liquid-phase sterilization, <1229.11> Vapor phase sterilization, in: United States Pharmacopeia 41, Vol. 5, United Book Press, Baltimore, 2018, pp. 7689–7706, 7709-7716, 7719-7722, 7733-7734.

[34] The European Pharmacopoeia Commission, 5.1.1 Methods of preparation of sterile products, in: European Pharmacopoeia 10th Edition, Druckerei C.H. Beck, Nördlingen, 2020, pp. 619–622.

[35] The committee on Japanese pharmacopoeia, G4 microorganisms: sterilization and sterilization indicators, in: Japanese Pharmacopoeia 17th Edition, Ministry of Health, Labour and Welfare, Tokyo, 2016, pp. 2507–2513.

[36] Sterilization of health care products — radiation — Part 1: Requirements for development, validation and routine control of a sterilization process for medical devices, ISO International Standard 11137–1 (2006).

[37] A. Koryakina, E. Frey, P. Bruegger, Cryopreservation of human monocytes for pharmacopeial monocyte activation test, J. Immunol. Methods 405 (2014) 181–191.

[38] K.A. Mattos, E.C.A. Navega, V.F. Silva, et al., Applicability of the monocyte activation test (MAT) in the quality control of the 17DD yellow fever vaccine, Altern. Lab. Anim. 46 (2018) 23–37.

[39] C. Wunderlich, S. Schumacher, M. Kietzmann, Pyrogen detection methods: comparison of bovine whole blood assay (bWBA) and monocyte activation test (MAT), BMC Pharmacol. Toxicol. 15 (2014), 50.

[40] S. Valentini, G. Santoro, F. Baffetta, et al., Monocyte-activation test to reliably measure the pyrogenic content of a vaccine: an in vitro pyrogen test to overcome in vivo limitations, Vaccine 37 (2019) 3754–3760.

[41] C.L.A. Utescher, K.L. Buosi, V.F. Botosso, et al., Monocyte activation test (MAT) as a possibility of replacement for the rabbit pyrogen test in hyperimmune sera, Braz. J. Pharm. Sci. 54 (2018), e17530.

[42] The European Pharmacopoeia Commission, 2.6.30 monocyte-activation test, in: European Pharmacopoeia 10th Edition, Druckerei C.H. Beck, Nördlingen, 2020, pp. 233–239.

[43] The United States Pharmacopeial Convention, 1226> Verification of compendial procedures, <1224> Transfer of analytical procedures, in: United States Pharmacopeia 41, Vol. 5, United Book Press, Baltimore, 2018, pp. 7671–7672, 7663-7665.

[44] J. Eichhorn, M. Mentgen-Wolny, Verification of pharmacopoeial methods in the analytical laboratory: how comprehensive in practice? Pharm. Ind. (Pharmind) 79 (2017) 274–279.

[45] T.T. Zuo, H.Y. Jin, M.Z. Xu, et al., Transfer, validation and verification of analytical procedures described in General chapter <1224> <1225> <1226> of U.S. Pharmacopoeia (USP37-NF32) and their significances for trace analysis assurance system of traditional Chinese medicines, Chin. J. Pharm. Anal. 36 (2016) 868–872.

[46] J. Ermer, M. Limberger, K. Lis, et al., The transfer of analytical procedures, J. Pharmaceut. Biomed. Anal. 85 (2013) 262–276.

[47] C. Fromke, L.A. Hothorn, F. Sczesny, et al., Analytical method transfer: improving interpretability with ratio-based statistical approaches, J. Pharmaceut. Biomed. Anal. 74 (2013) 186–193.

[48] L. Kaminski, U. Schepers, H. Watzig, Analytical method transfer using equivalence tests with reasonable acceptance criteria and appropriate effort: extension of the ISPE concept, J. Pharmaceut. Biomed. Anal. 53 (2010) 1124–1129.

[49] C. Agut, A. Caron, C. Giordano, et al., Transfer of analytical procedures: a panel of strategies selected for risk management with emphasis on an integrated equivalence-based comparative testing approach, J. Pharmaceut. Biomed. Anal. 56 (2011) 293–303.

[50] N.L. Vu, K. Nguyen, T. Kupiec, The essentials of United States Pharmacopoeia Chapter antimicrobial effectiveness testing and its application in pharmaceutical compounding, Int. J. Pharm. Compd. 18 (2014) 123–130.

[51] H.R. He, F. Gao, Y.H. Zhang, et al., Effect of processing on the reduction of pesticide residues in a traditional Chinese medicine (TCM), Food Addit. Contam. 37 (2020) 1156–1164.

[52] J.W. Huang, T.G. Nan, Y. Yuan, et al., Basic data investigation for quality of traditional Chinese medicine analysis of pesticide residues, Chin. J. Exp. Tradit. Med. Formulae. 24 (2017) 56–61.

[53] J.J. Xiao, X. Xu, W. Fan, et al., Analysis of exposure to pesticide residues from traditional Chinese medicine, J. Hazard Mater. 365 (2019) 857–867.

[54] Y.Y. Zhang, G.M. Han, C. Sun, et al., Research on sample preparation

中国药材中农药残留的检测技术, *Res. Pract. Chin. Med.* 3 (2015) 81–84. 王丽丽、孔维军、杨明华等, 中国传统药材的安全问题及新型快速检测方法, *Acta Pharm. Sin.* B. 5 (2015) 38–46. 刘正凤、徐俊、李建明等, 灵芝中农药残留的GC-MS及GC-MS/MS测定, *Res. Pract. Chin. Med.* 5 (2015) 20–23. 谢辉、张旭、王新等, 抗黄曲霉毒素B单克隆抗体的制备及其在间接竞争ELISA检测黄曲霉毒素B中的应用, *Microbiology (Beijing, China)* 42 (2015) 2033–2040. 南挺刚、洪晓燕、徐霞颖等, 中药材黄曲霉毒素酶联免疫吸附测定法的开发, *China J. Chin. Mater. Med.* 45 (2020) 4158–4162. Roman B.E., Driksna D., Abouzied M.M.等, Veratox总黄曲霉毒素ELISA试剂盒的MAX水提方法验证, *J. AOAC Int.* 100 (2017) 1131–1133

Technol. 22 (2017) 130e135. [76] R. Medina, C.A. Garcia, M. Hurtado 等, 关于USP拨片与流通细胞溶解方法的比较, 用于测试固定剂量组合配方中酮丙醇和乙酰氨基酚的检测方法, 《拉丁美洲药理学杂志》35 (2016) 1573e1581. [77] A. Paprská'rov'a, P. Mo'zn'a, E.F. Oga等, 《流通式USP IV溶解装置用于评估低溶解性基本药品的仪器化: 技术说明》, *AAPS制药科学技术* 17卷 (2016年) 1261e1266. [78] S. Qiu, K. Wang, M.Z. Li 等, 利用流穿细胞应用 4, 即释和缓释制剂的体外溶解研究, *Dissolution Technol.* 21 (2014) 6e16. [79] L.H. Emara, E.W. Elsayed, A.A. El-Ashmawy 等, 《流动细胞作为体外溶解鉴别工具用于评估格列齐德固体分散物》, 《药学应用杂志》。《科学》7卷 (2017年) 70卷第777页. [80] N.D. Rudd, M. Reibarkh, R. Fang 等, 体外释放解读

techniques for pesticide residues in Chinese medicinal materials, *Res. Pract. Chin. Med.* 3 (2015) 81–84. [55] L.L. Wang, W.J. Kong, M.H. Yang, et al., Safety issues and new rapid detection methods in traditional Chinese medicinal materials, *Acta Pharm. Sin. B.* 5 (2015) 38–46. [56] Z.F. Liu, J. Xu, J.M. Li, et al., Determination of pesticide residues in ganoderma lucidum by GC-MS and GC-MS/MS, *Res. Pract. Chin. Med.* 5 (2015) 20–23. [57] H. Xie, X. Zhang, X. Wang, et al., Preparation of anti-aflatoxin B₁ monoclonal antibodies and its use in an indirect competitive ELISA for aflatoxin B₁, *Microbiology (Beijing, China)* 42 (2015) 2033–2040. [58] T.G. Nan, X.Y. Hong, X.Y. Xu, et al., Development of the enzyme-linked immunosorbent assay of aflatoxin of Chinese herbal medicines, *China J. Chin. Mater. Med.* 45 (2020) 4158–4162. [59] B.E. Roman, D. Driksna, M.M. Abouzied, et al., Validation of MAX aqueous extraction on Veratox® for total aflatoxin ELISA test kit, *J. AOAC Int.* 100 (2017) 1131–1133. [60] H. Peng, Y.W. Chang, R.C. Baker, et al., Interference of mycotoxin binders with ELISA, HPLC and LC-MS/MS analysis of aflatoxins in maize and maize gluten, *Food Addit. Contam.* 37 (2020) 496–506. [61] C.S. Pereira, S.C. Cunha, J.O. Fernandes, Validation of an enzyme-linked immunosorbent assay (ELISA) test kit for determination of aflatoxin B₁ in corn feed and comparison with liquid-chromatography tandem mass spectrometry (LC-MS/MS) Method, *Food Anal. Method.* 13 (2020) 1806–1816. [62] A.M. Beyene, X.W. Du, D.E. Schrunck, et al., High-performance liquid chromatography and enzyme-linked immunosorbent assay techniques for detection and quantification of aflatoxin B₁ in feed samples: a comparative study, *BMC Res. Notes* 12 (2019), 492. [63] T. Yamasaki, S. Miyake, N. Sato, et al., Development of enzyme-linked immunosorbent assay for analysis of total aflatoxins based on monoclonal antibody reactive with aflatoxins B₁, B₂, G₁ and G₂, *J. Food Hyg. Soc. Jpn.* 59 (2018) 200–205. [64] Evaluation and recommendation of pharmacopoeial texts for use in the ICH regions Q4B, Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). <https://www.ich.org/page/quality-guidelines>. (accessed on 11 December, 2020). [65] X.X. Xu, H.Y. Xu, G.M. Jin, et al., Overview and analysis of general chapters (appendices) of physical and chemical testing methods in pharmacopoeias, *Chin. Pharmaceut. J.* 15 (2018) 1323–1332. [66] Z. Zhang, X.X. Xu, Z. Liu, et al., Comparative assessment of the differences between Chinese Pharmacopoeia and ICH Q4B detection methods, *Chin. Food & Drug Admin. Mag.* 12 (2019) 24–33. [67] X.X. Xu, Z. Liu, Z. Zhang, et al., Review of the Chinese Pharmacopoeia tablet friability test revision history and prospects for harmonization with ICH, *Chin. Pharm.* 22 (2019) 1138–1140. [68] Stability testing of new drug substances and products Q1A (R2), Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). <https://database.ich.org/sites/default/files/Q1A%28R2%29%20Guideline.pdf>. (accessed on 11 December, 2020). [69] Validation of analytical procedures: text and methodology Q2 (R1), Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). https://database.ich.org/sites/default/files/Q2_R1_Guideline.pdf. (accessed on 11 December, 2020). [70] Impurities in new drug substances Q3A (R2), Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). https://database.ich.org/sites/default/files/Q3A_R2_Guideline.pdf. (accessed on 11 December, 2020). [71] Impurities in new drug products Q3B (R2), Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). https://database.ich.org/sites/default/files/Q3B_R2_Guideline.pdf. (accessed on 11 December, 2020). [72] Impurities for residual solvents Q3C (R6), Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). https://database.ich.org/sites/default/files/Q3C-R6_Guideline_ErrorCorrection_2019_0410_0.pdf. (accessed on 11 December, 2020). [73] Evaluation and recommendation of pharmacopoeial texts for use in the ICH regions on dissolution test general chapter Q4B Annex 7 (R2), Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). <https://database.ich.org/sites/default/files/Q4B%20Annex%207%20%28R2%29%20Guideline.pdf>. (accessed on 11 December, 2020). [74] The European Pharmacopoeia Commission, 2.9.3 Dissolution test for solid dosage forms, in: *European Pharmacopoeia 10th Edition*, Druckerei C.H. Beck, Nördlingen, 2020, pp. 326–333. [75] S. Hori, T. Kawada, S. Kogure, et al., Comparative release studies on suppositories using the basket, paddle, dialysis tubing and flow-through cell methods I. Acetaminophen in a lipophilic base suppository, *Pharmaceut. Dev.*

Technol. 22 (2017) 130–135. [76] R. Medina, C.A. Garcia, M. Hurtado, et al., Comparison of USP paddle and flow-through cell dissolution methods for testing ketoprofen and acetaminophen from fixed-dose combination formulation, *Lat. Am. J. Pharm.* 35 (2016) 1573–1581. [77] A. Paprskářová, P. Možná, E.F. Oga, et al., Instrumentation of flow-through USP IV dissolution apparatus to assess poorly soluble basic drug products: a technical note, *AAPS PharmSciTech* 17 (2016) 1261–1266. [78] S. Qiu, K. Wang, M.Z. Li, et al., In vitro dissolution studies of immediate-release and extended-release formulations using flow-through cell apparatus 4, *Dissolution Technol.* 21 (2014) 6–16. [79] L.H. Emara, E.W. Elsayed, A.A. El-Ashmawy, et al., The flow-through cell as an in vitro dissolution discriminative tool for evaluation of gliclazide solid dispersions, *J. Appl. Pharmaceut. Sci.* 7 (2017) 70–77. [80] N.D. Rudd, M. Reibarkh, R. Fang, et al., Interpreting in vitro release performance from long-acting parenteral nanosuspensions using USP-4 dissolution and spectroscopic techniques, *Mol. Pharm.* 17 (2020) 1734–1747. [81] B.R. Pezzini, M.G. Issa, M.D. Duque, et al., Applications of USP apparatus 3 in assessing the in vitro release of solid oral dosage forms, *Braz. J. Pharm. Sci.* 51 (2015) 265–273. [82] S. Perivilli, M. Kakhi, E. Stippler, Computational fluid dynamics simulation of hydrodynamics in USP apparatus 3-the influence of dip rate, *Pharm. Res.* 32 (2015) 1304–1315. [83] Evaluation and recommendation of pharmacopoeial texts for use in the ICH regions on sterility test general Chapter Q4B Annex 8 (R1), Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). <https://database.ich.org/sites/default/files/Q4B%20Annex%208%28R1%29%20Guideline.pdf>. (accessed on 11 December, 2020). [84] Evaluation and recommendation of pharmacopoeial texts for use in the ICH regions on bulk density and tapped density of powders general Chapter Q4B Annex 13, Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). https://database.ich.org/sites/default/files/Q4B_Annex_13_Annex.pdf. (accessed on 11 December, 2020). [85] J.P.S. Silva, D. Splendor, I.M.B. Goncalves, et al., Note on the measurement of bulk density and tapped density of powders according to the European Pharmacopoeia, *AAPS PharmSciTech* 14 (2013) 1098–1100. [86] I. Akseli, J. Hilden, J.M. katz, et al., Reproducibility of the measurement of bulk/tapped density of pharmaceutical powders between pharmaceutical laboratories, *J. Pharm. Sci.* 108 (2019) 1081–1084. [87] Assessment and control of DNA reactive (mutagenic) impurities in pharmaceuticals to limit potential carcinogenic risk M7 (R1), Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). https://database.ich.org/sites/default/files/M7_R1_Guideline.pdf. (accessed on 11 December, 2020). [88] P. Nethercote, J. Ermer, Quality by design for analytical methods: implications for method validation and transfer, *Pharmaceut. Technol.* 36 (2012) 74–79. [89] Pharmaceutical development Q8 (R2), Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). <https://database.ich.org/sites/default/files/Q8%28R2%29%20Guideline.pdf>. (accessed on 11 December, 2020). [90] Quality risk management Q9, Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). <https://database.ich.org/sites/default/files/Q9%20Guideline.pdf>. (accessed on 11 December, 2020). [91] Pharmaceutical quality system Q10, Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). <https://database.ich.org/sites/default/files/Q10%20Guideline.pdf>. (accessed on 11 December, 2020). [92] Development and manufacture of drug substances (chemical entities and biotechnological/biological entities) Q11, Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). <https://database.ich.org/sites/default/files/Q11%20Guideline.pdf>. (accessed on 11 December, 2020). [93] Technical and regulatory considerations for pharmaceutical product lifecycle management Q12, Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). https://database.ich.org/sites/default/files/Q12_Guideline_Step4_2019_1119.pdf. (accessed on 11 December, 2020). [94] Ich Q13: Continuous manufacturing for drug substances and drug products, compilation prepared by international council for harmonisation of technical requirements for pharmaceuticals for human use (ICH). <https://database.ich.org/sites/default/files/Q13%20Business%20Plan.pdf>. (accessed on 11 December, 2020). [95] Ich Q14: Analytical procedure development and revision of Q2 (R1) analytical validation, Compilation prepared by International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). https://database.ich.org/sites/default/files/Q2R2-Q14_EWG_Concept_

Paper.pdf. (访问日期: 2020年12月11日)。[96] M.K. Parr, A.H. Schmidt, 分析方法的生命周期管理, 《药物与生物分析杂志》147 (2018) 506-517。[97] A. Bugno, D.P.S. Sanches, A.A.B. Almodovar 等, 快速无菌检测方法与药典无菌检测的性能调查与比较, 《药物创新杂志》13 (2018) 27-35。[98] M.J. Miller, E.V. Heuvel, D. Roesti, 统计分析在验证快速微生物方法中的作用, 《欧洲生物药学评论》21 (2016) 46-53。[99] J. Chisholm, S. Bhatt, A. Chaboureau 等, 简化策略用于在…

商业化快速微生物学方法 (RMM) 用于加拿大监管审批的室内验证, Cytotherapy 19 (2017) 1529–1536。S. Suessner, S. Hennerbichler, S. Schreiberhuber 等, 组织产品替代微生物学方法的验证, Cell Tissue Bank. 15 (2014) 277–286。J. Moldenhauer, 向快速微生物学方法的争相采用——或不, Eur. Pharm. Rev. 22 (2017) 28–30。

- [Paper.pdf](#). (accessed on 11 December, 2020).
- [96] M.K. Parr, A.H. Schmidt, Life cycle management of analytical methods, J. Pharmaceut. Biomed. Anal. 147 (2018) 506–517.
- [97] A. Bugno, D.P.S. Sanches, A.A.B. Almodovar, et al., Performance survey and comparison between rapid sterility testing method and pharmacopoeia sterility test, J. Pharm. Inno. 13 (2018) 27–35.
- [98] M.J. Miller, E.V. Heuvel, D. Roesti, The role of statistical analysis in validating rapid microbiological methods, Eur. Biopharm. Rev. 21 (2016) 46–53.
- [99] J. Chisholm, S. Bhatt, A. Chaboureau, et al., Strategy for an abbreviated in-

- house qualification of a commercially available Rapid Microbiology Method (RMM) for canadian regulatory approval, Cytotherapy 19 (2017) 1529–1536.
- [100] S. Suessner, S. Hennerbichler, S. Schreiberhuber, et al., Validation of an alternative microbiological method for tissue products, Cell Tissue Bank. 15 (2014) 277–286.
- [101] J. Moldenhauer, The rush to rapid microbiological methods-or not, Eur. Pharm. Rev. 22 (2017) 28–30.